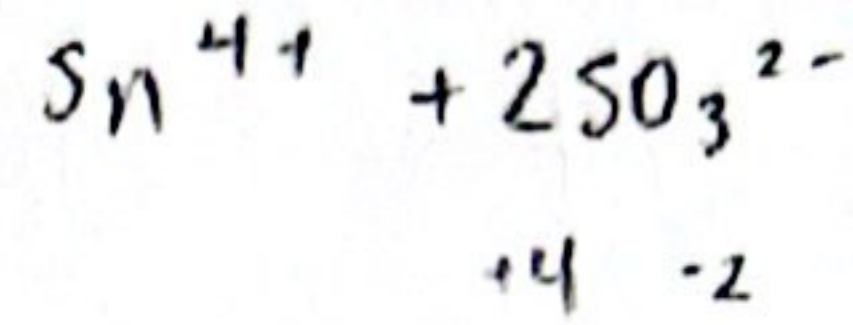


Chapter 9 Assessment

1. Determine the oxidation number of each element in $\text{Sn}(\text{SO}_3)_2$ (3 marks)

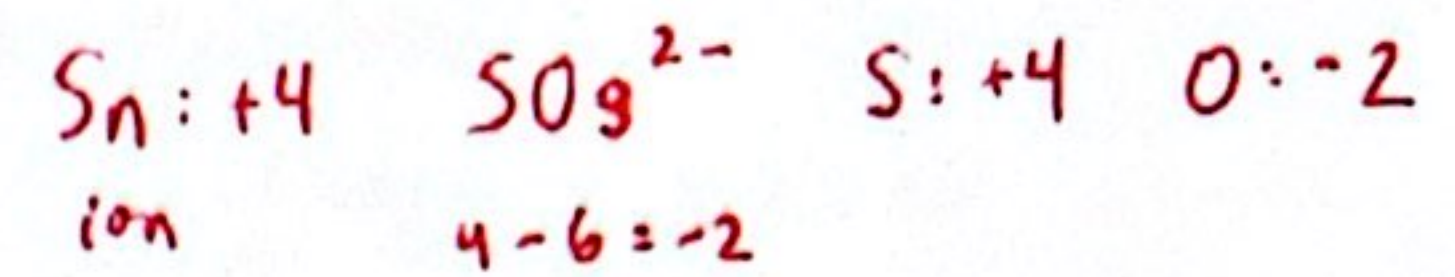
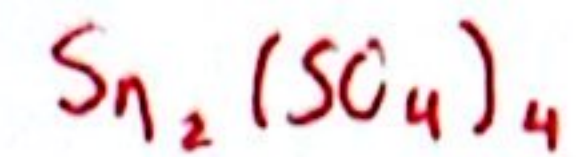
	Oxidation number
Sn	+4 ✓
S	+4 ✓
O	-2 ✓



$$-2 = 3(-2) + x$$

$$x = +4$$

when you do backwards crisscross

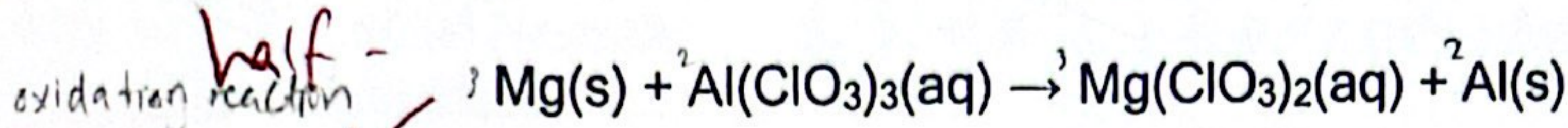


9.3 not on exam (#4).

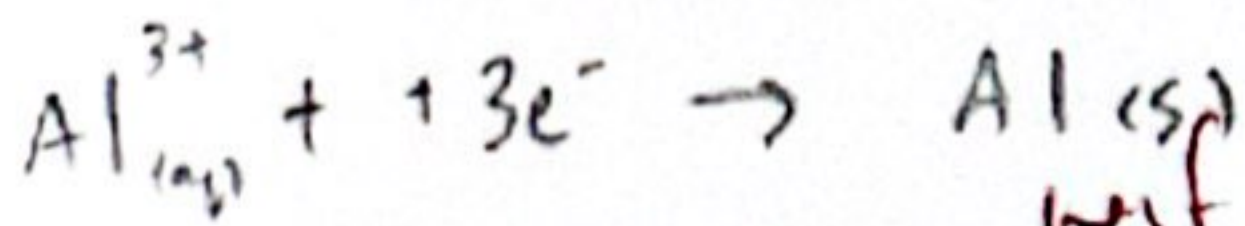
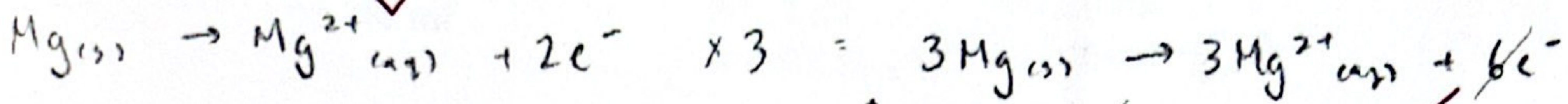
Q1-2 are 9.1

so can be on MC

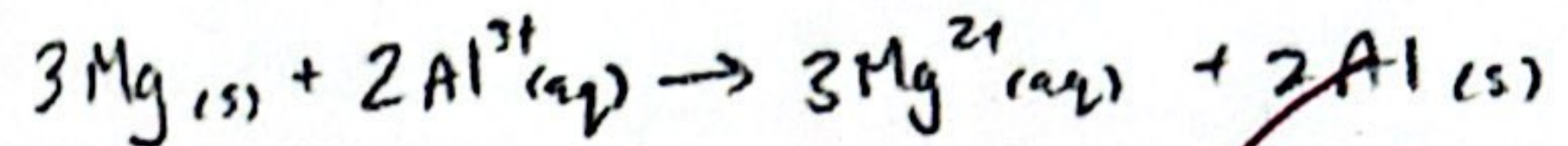
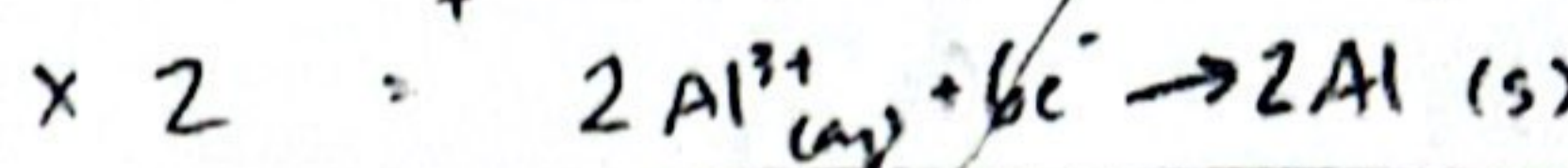
2. For the following reaction, write the 2 labelled and balanced half reactions and a net reaction: (4 marks)



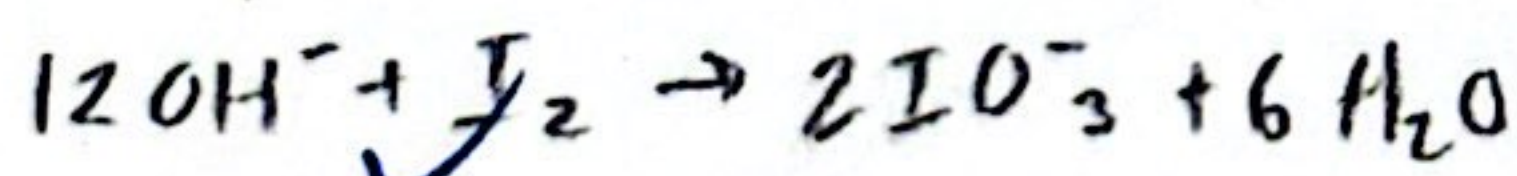
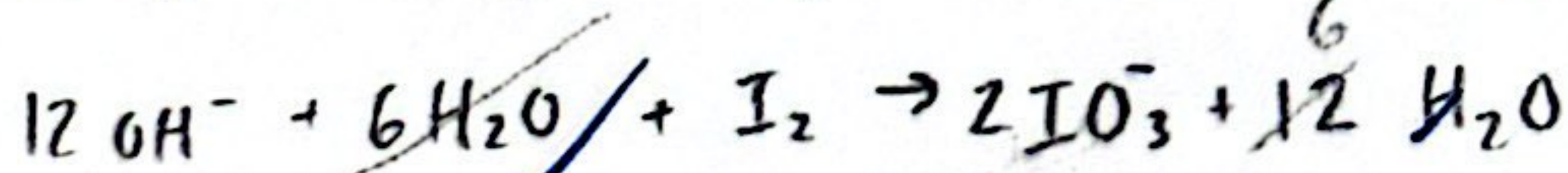
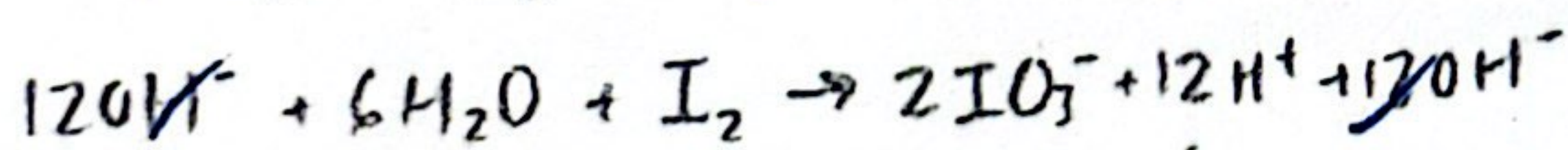
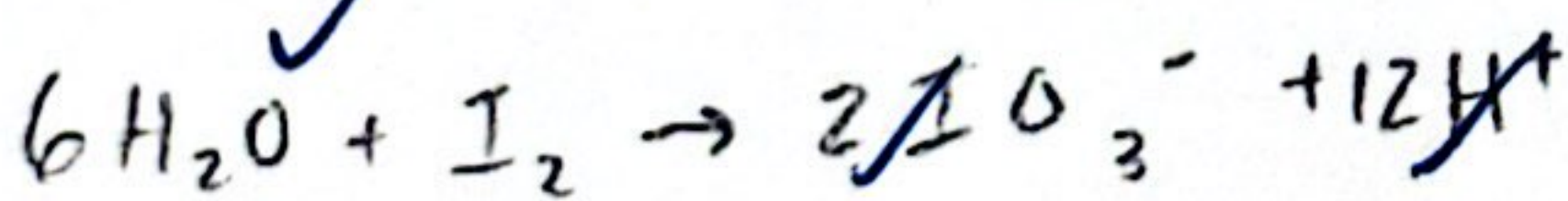
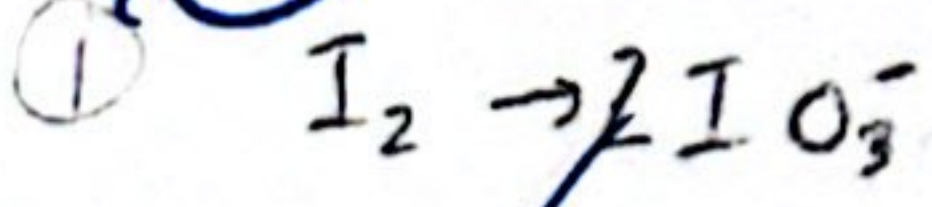
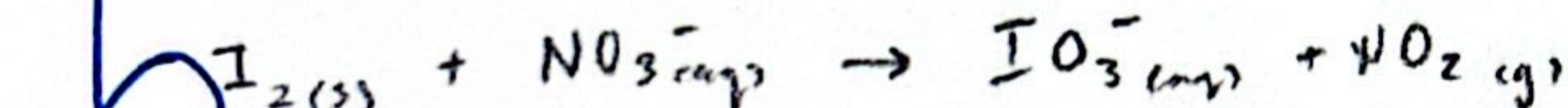
oxidation half-reaction



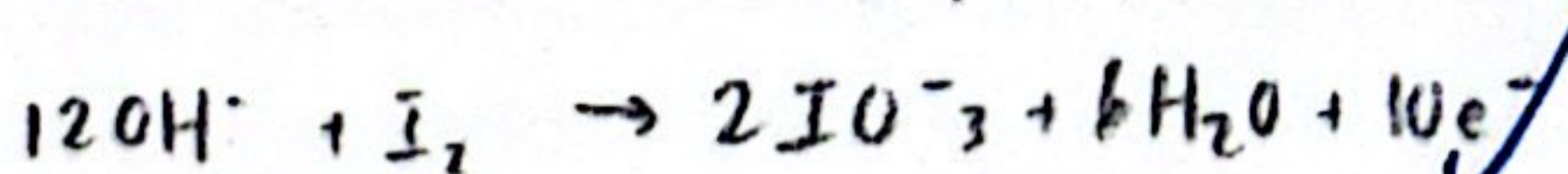
reduction half-reaction



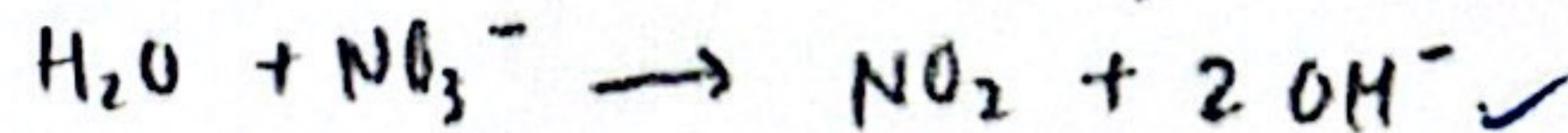
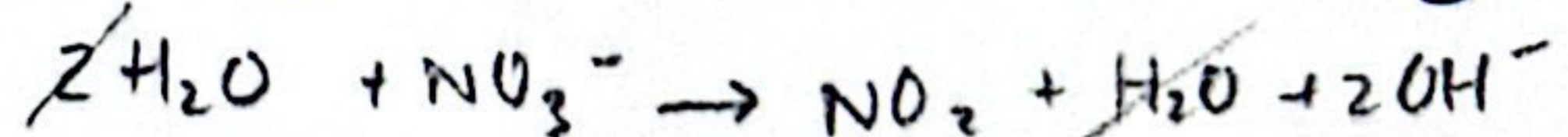
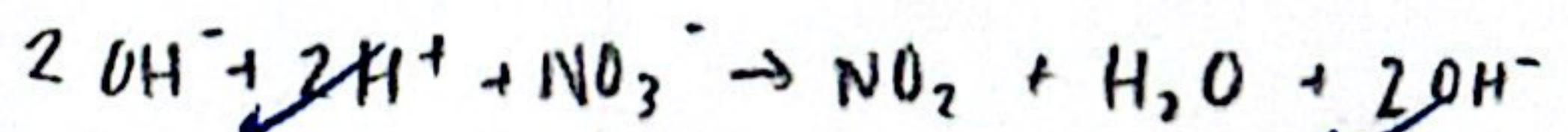
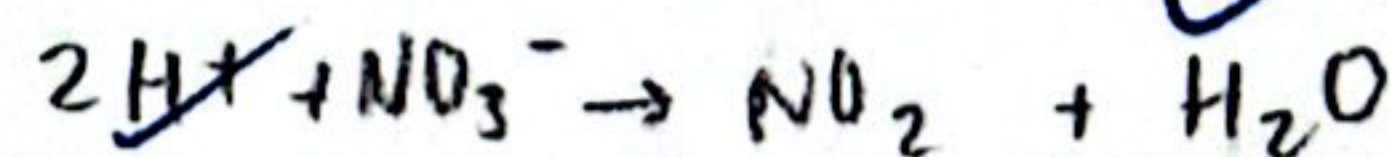
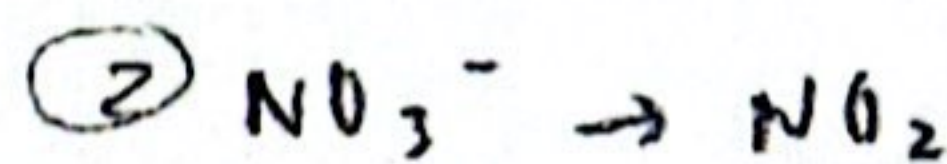
* know how to do both for exam

3. Balance the following redox reaction in a *basic* solution. Use either the half reaction method or the oxidation number method in your solution: $\text{I}_2(\text{s}) + \text{NO}_3^-(\text{aq}) \rightarrow \text{IO}_3^-(\text{aq}) + \text{NO}_2(\text{g})$ (6 marks)

$$12(-1) = -12 \quad 2(-1) + 6(0) = -2$$

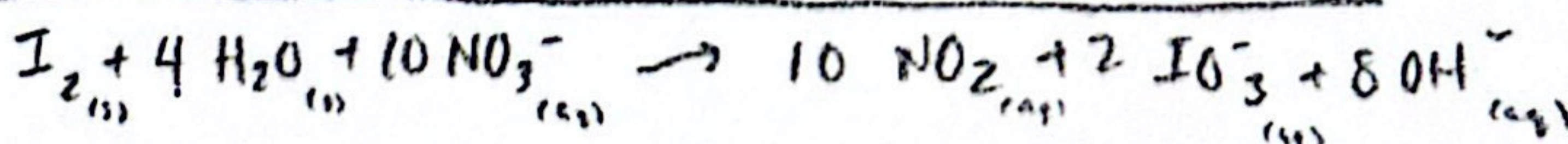
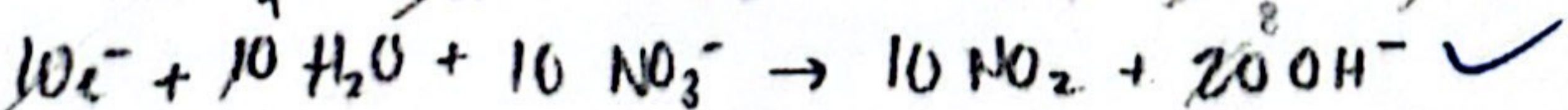
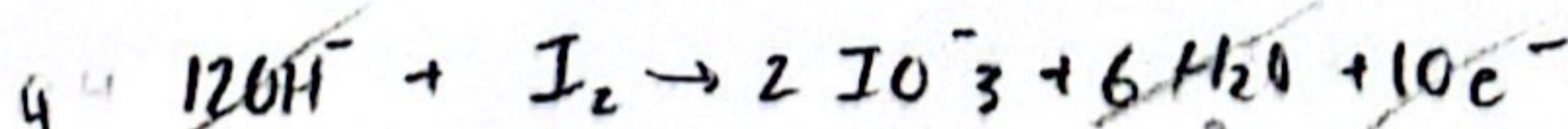
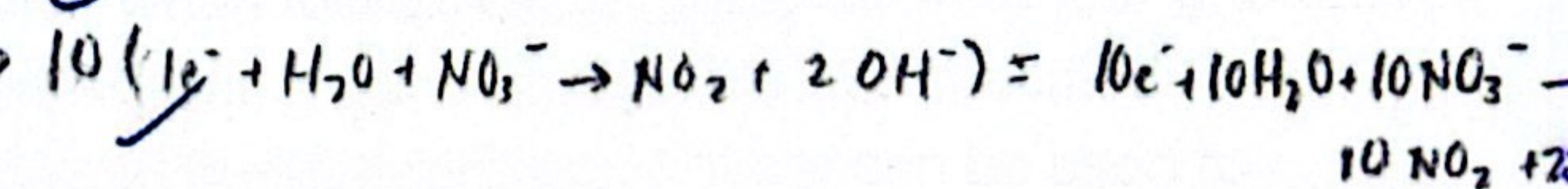
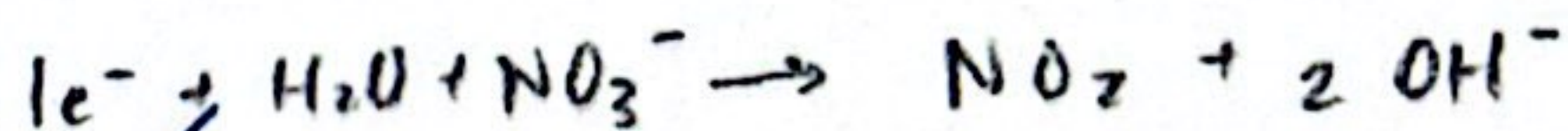


(1) + (2)

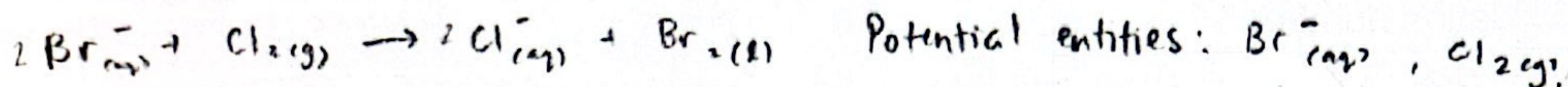


$$1(-1) = -1$$

$$2(-1) = -2$$



4. Determine whether the following redox reaction will occur spontaneously. Refer to the Table of Relative Strengths of Oxidizing Agents and Reducing Agents: bromide ions and chlorine gas to produce bromine ^{liquid} gas and chloride ions. Make sure to write a net ionic equation and identify the oxidizing agent and reducing agent. (3 marks)



Potential entities: $\text{Br}^{-}(\text{aq})$, $\text{Cl}_2(\text{g})$

↓ ↓
SRA SOA

The strong oxidizing agent is higher on the table than the strong reducing agent

DA \ RA

The reaction is spontaneous ✓

$\text{Br}^{-}(\text{aq})$ is the reducing agent
 $\text{Cl}_2(\text{g})$ is the oxidizing agent

From the table: $2\text{Br}^{-}(\text{aq}) \rightleftharpoons 2\text{e}^{-} + \text{Br}_2(\text{l})$

